

**Fifth Semester B. Tech. (Electronics and Communication Engineering)
Examination**

DATABASE MANAGEMENT SYSTEM

Time : 3 Hours]

[Max. Marks : 60

Instructions to Candidates :—

All questions are compulsory.

1. (a) Enlist and explain various types of architecture. 5(CO1)
 (b) For the given relations R and S,

R

SNo	Dept
S1	Phy
S2	Psy
S3	Chem
S4	Bio

S

SNo	Dept
S10	Maths1
S3	Chem
S15	Eng
S16	Maths2

Evaluate :

- (i) $R \cup S$.
 (ii) $R \cap S$.
 (iii) $S - R$.
 (iv) $R \bowtie_{\text{Dept}=\text{Chem}} S$. 5(CO1)

2. (a) What is ER Model of database ? Explain various components of ER Model. 5(CO4)

- (b) For the following Functional Dependency (FD) sets F1 and F2, check whether F1 and F2 are equivalent or not.
 $R = (A, C, D, E, H)$
 $F1 = \{A \rightarrow C, AC \rightarrow D, E \rightarrow AD\}$
 $F2 = \{A \rightarrow CD, E \rightarrow AH\}$ 5(CO1)
3. Design the database with the following schema :
 Branch(b_name, assets, b_city)
 Account(acc_no, balance, b_name)
 Depositor(c_name, acc_no)
 Customer(c_name, c_address, c_city)
 Barrower(c_name, I_no)
 Loan(I_no, amount, b_name)
 Write the SQL query to :
- (i) Find the name of all the customers having bank account.
 - (ii) Find branch name and account number where balance is ≥ 100 and $\leq 1,000$.
 - (iii) Find the name of the customers having account in "Katol road" branch.
 - (iv) Find loan number and amount of loan from the branch situated in "Sitabuldi".
 - (v) Find customer names which have loan from branch having assets $\leq 1,00,000$. 10(CO3,5)
4. (a) Explain the need for indexing. Also explain various types of indexing. 5(CO2)
- (b) Consider a hash table of size seven, with starting index zero and a hash function, $h(K) = (3K + 4)$. Assuming the hash table is initially empty, determine the contents of the table when the sequence 1, 3, 8, 10 is inserted into the table using closed hashing. 5(CO1)
5. (a) What is query processing ? Explain the steps involved in query processing and explain it in detail. 5(CO1)
- (b) Explain ACID property of transaction in Database. 5(CO2)

6. (a) Consider the following schedules :

$S_1 : r_1(x), r_1(y), r_2(x), r_2(y), w_2(y), w_1(x)$

$S_2 : r_1(x), r_2(x), r_2(y), w_2(y), r_1(y), w_1(x)$

Prove that S_1 is not a conflict serializable and S_2 is conflict serializable.
Given that, $r_i(z)$ and $w_i(z)$ denote read and write operations respectively
on data item Z by a transaction T_i . 5(CO5)

- (b) What is the two-phase locking protocol ? Two-phase locking protocol guarantees Serializability. Justify. 5(CO4)

